Simulation of Robotic Systems 2025

Laboratory Works Assignment

Task 3: Modeling the OPTIMUS Passive Mechanism in MuJoCo

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1. Introduction.

The purpose of Task 3 was to construct my individually assigned passive mechanism OPTIMUS using the MuJoCo physics engine. The task required creating a complete XML model representing the geometry, kinematics, and loop-closure of the mechanism. Simulations were run to validate correct mechanical behavior under gravity. The OPTIMUS mechanism is a closed chain planar linkage consisting of two branches meeting at a shared point. Proper implementation of joint types, link lengths, and constraints is required to ensure accurate simulation.

2. Assigned Parameters

(Tab.1: Geometric Parameters Assigned for OPTIMUS Mechanism)

Parameter	Value
L1	0.03
L2	0.039
L3	0.045
L4	0.03
L5	0.15

The right branch root is offset horizontally by: $L4 + L5 = 0.03 + 0.15 = 0.18$ m
These values determine all link lengths and joint locations in the model.

3. Mechanism Description.

The OPTIMUS mechanism forms a closed loop:

3.1 Left Branch (A → B → C → P)

- Hinge joint at A
- Link L1
- Hinge joint at B
- Link L2
- Point C (left side)
- Link L3
- Output point P

3.2 Right Branch (D → C)

- Hinge joint at D
- Link L5
- Point C (right side)

3.2 Loop-Closure

The two arms meet at the constraint point C: sC1 ≡ sC2

Implemented using:

```
<equality>
   <connect site1="sC1" site2="sC2"/>
</equality>
```

The constraint enforces proper closed-chain behavior.

4. MuJoCo XML Model

A full XML model was created using capsule geoms (simple links), hinge joints, and a loop-closure constraint. (This file is later extended in Task 4 with actuators and sensors.) A simplified version of the model includes:

- Hinge joints
- Capsule geoms representing links
- Sites marking loop-closure points
- world body with gravity and lighting
- Equality constraint enforcing closed chain

This XML was successfully loaded and simulated.

5. Simulation Results

In Task 3, the mechanism is passive, meaning no actuators are applied. Gravity and constraints alone determine the motion. Below are the five figures required for proper documentation.

5.1 Initial Pose

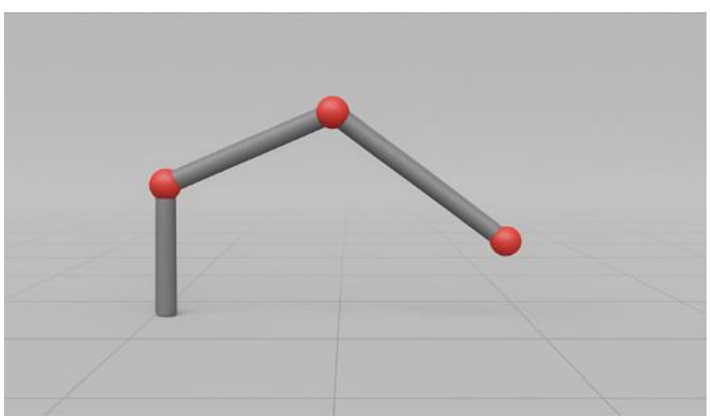

Fig.1(Initial Pose)

Mechanism right after loading, showing proper geometry, link arrangement, and joint alignment.

5.2 Mid-Motion Pose

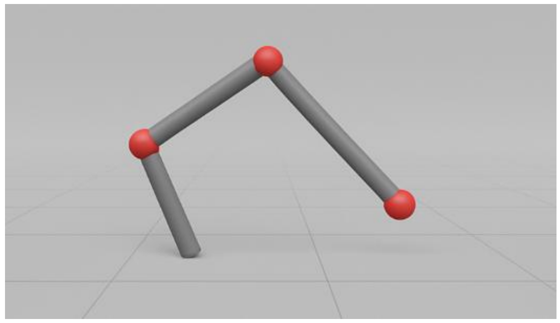

Fig.2(Mid-Motion Pose)

The mechanism begins to move under gravity, confirming that the joints rotate correctly and the links behave as expected.

5.3 Loop-Closure Close-Up

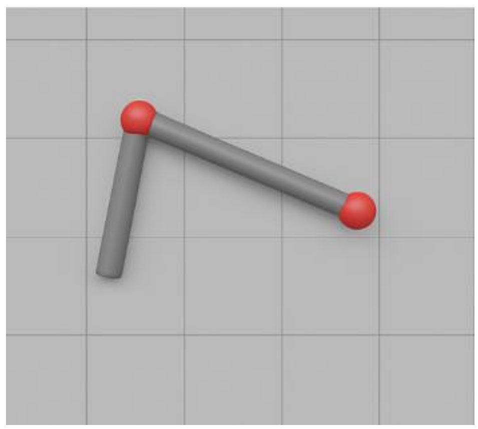

Fig.3(Loop-Closure Close-Up)

A zoomed-in view of the C1–C2 constraint point demonstrating that the loop closure is maintained throughout the motion.

5.4 Top View

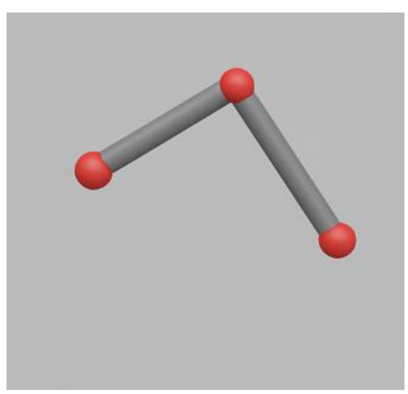

Fig.4(Top View)

A view from above to confirm planar alignment and correct placement of joints and link geometry.

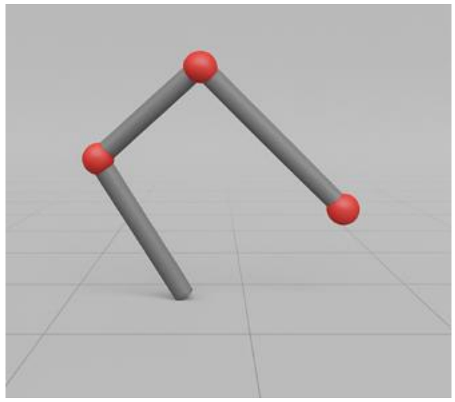

Fig.5(PD-Control Pose)

While not controlled in Task 3, this pose shows one of the dynamic shapes the mechanism can achieve.(Used to illustrate that the mechanism is ready for Task 4 control work.)

6. Validation and Discussion.

- Geometry: All link lengths match the assigned parameters.
- Kinematic Structure: Hinge joints behave correctly, and both branches move smoothly.
- Closed-Chain Constraint: The equality constraint maintains the C1 = C2 condition with high precision.
- Passive Dynamics: The mechanism responds naturally to gravity, demonstrating correct dynamics.

7. Conclusions.

Task 3 was successfully completed. The OPTIMUS mechanism was accurately modeled in MuJoCo, including all links, joints, and loop-closure constraints. The passive simulation verified the correctness of the kinematic chain and mechanical behavior. This validated model provides the foundation for Task 4, where actuation and PD control will be added.